CSE470: Software Engineering

Incremental and Iterative Model

Evolutionary Process Model

- **Example**: Iterative, Incremental and Spiral model
- Can change requirements as you want
- Can go back to previous phases, such as after coding, we can go back to communication phase for requirement collection again.
- **Advantage**: Customer involvement, taking feedback from the customer during development make it correct, efficient and reliable, building the right product
- **Disadvantage**: There is a risk of the project expanding beyond its initial scope.

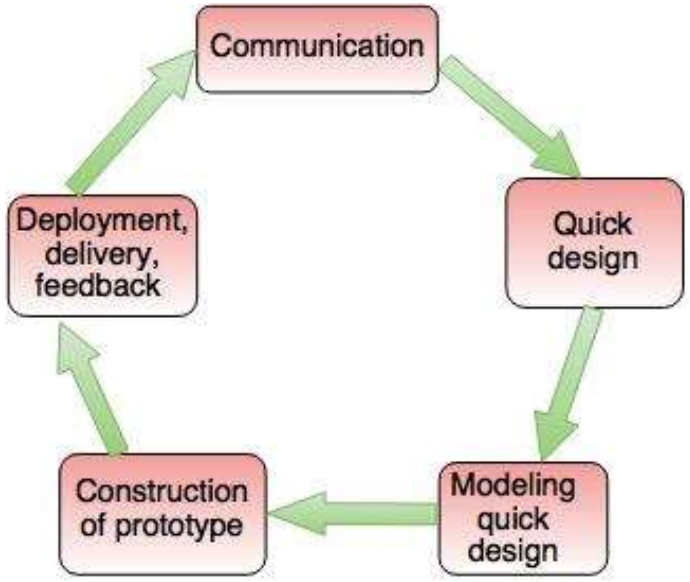


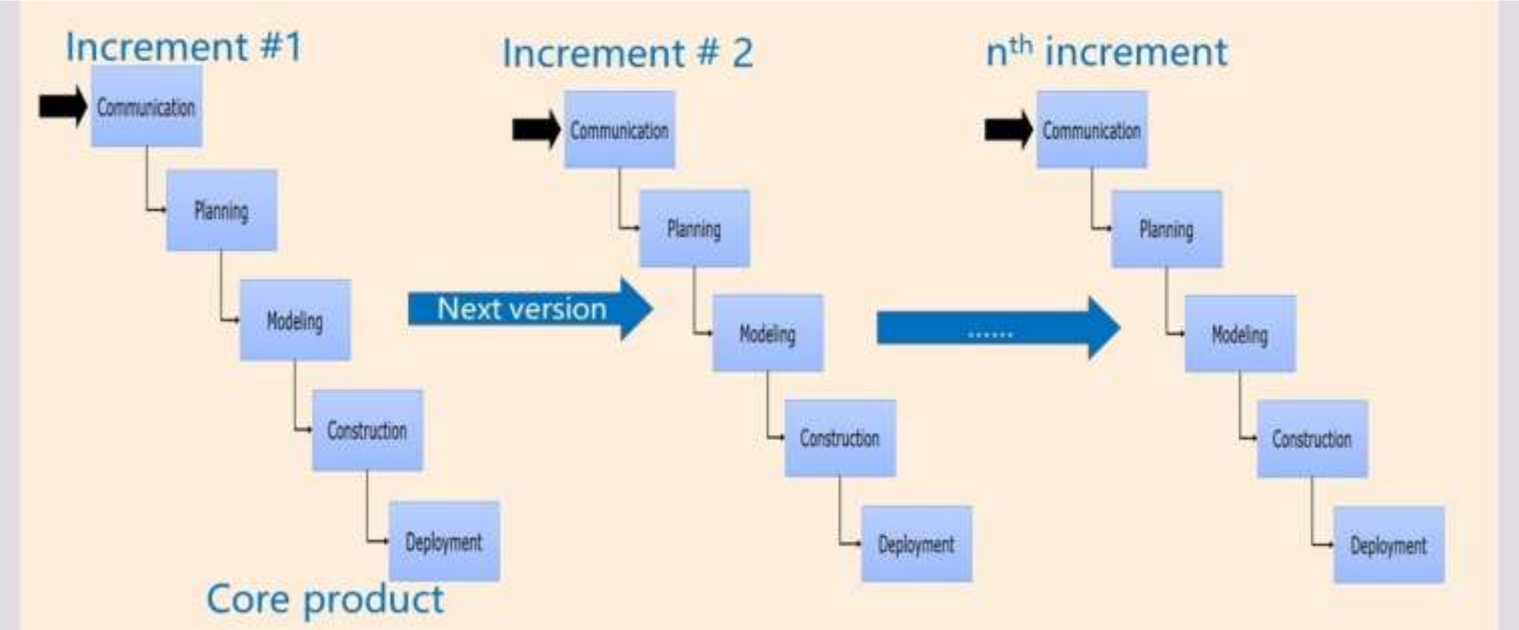
Fig. - The Prototyping Model

Incremental Model

A software process model where the software will be delivered in increments

- Customer wants to use the software from the very beginning of the project.
- Customer is quite well-known about the requirements (SRS is done and locked at the start of the each phase until development ends of that phase)
- The software is divided into **fixed number of increments** to be delivered to customers

Process of Incremental Model



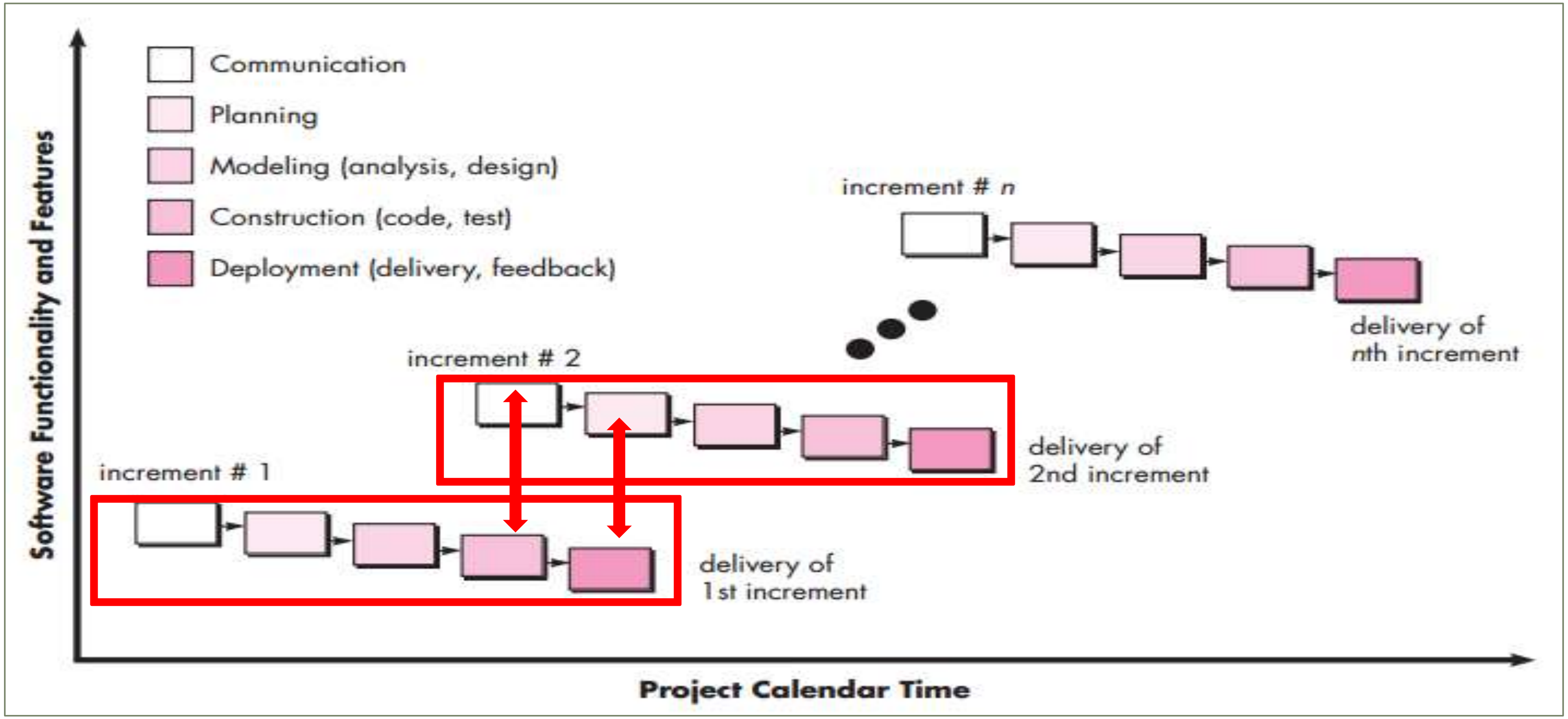
For example, in first increment Login module is delivered. In second increment, another new module such as Navigation module is also added. In third, one more added.

That is, this model “adds onto” new section as increments.

When to choose Incremental Model

- You are ready with thin slices of the overall software pieces.
- Customer wants **<u>prototype</u>** version of the software from the beginning of the project
- Parallel stages such as requirement collection, planning etc. can take place while construction and development phase.
- Increments needs to be prioritized by customers, so better chance of success (Customer feedback leads to a successful project)
- Better for small or medium size projects

Incremental Process Model



Iterative Model

A software process model where the software will be delivered in iterations

- Customer is not sure about the requirements
- Even development team may not be sure about which technology, algorithms may be used.
- There is no fixed limit of iterations, that is time is set aside.

Process of Iterative Model

For example, in first iteration Login module is delivered. In second iteration, Login module is updated again. In third iteration, Navigation module is added . And in fourth iteration some refinement is done.

That is, this model “changes or reworks” on same section in iterations until customer accepts it.

When to choose Iterative Model

1. Requirements are not fixed
2. Technological tools or requirements are not identified yet.
3. Instead of fixed time, quality of the features is refined with time
4. Customer feedbacks with repetitive iterations increase the product quality
5. Better for long-term and complex projects

Case Study

Being a project manager of a software company, you have got a project request for developing a coronavirus awareness app. The customers initially want the app to show testing info, take appointments and visualize affected area data. Currently, the software should support only Bangla language, however English language support will be added later.

In such a case, which would you apply – Waterfall, Incremental or Iterative model?

References

1. Roger S Pressman, Roger Pressman,"Software Engineering: A Practitioner's Approach", McGraw-Hill, 7th edition, 2010.
2. https://www.geeksforgeeks.org/software-engineering/software-engineering-incremental-process-model/